

GATE B PROTECTED REVIEW - Email Subject Line

RC2 | Recomputed controls | Threshold/CI distinction | State machine unchanged

Open only after a genuine saved Gate attempt. This review diagnoses the attempt; it must not be used as a pre-attempt solution.

Layer 1 - Data-integrity controls

Control	Expected
Rows / units	2000
Duplicate unit IDs	0
Missing unit IDs	0
Invalid group labels	0
Invalid converted labels	0
Invalid support-contact labels	0
Control n	1000
Treatment n	1000

Layer 2 - Statistical controls

Metric	Expected
Control conversions	100
Treatment conversions	105
Control rate	10.0000%
Treatment rate	10.5000%
Absolute lift	0.5000pp
Relative lift	5.0000%
z statistic	0.3686
two-sided p-value	0.7124
95% CI for rate difference	[-2.1585pp, 3.1585pp]
Support contacts	Control 55 Treatment 54
Support rates	5.5000% vs 5.4000%
Support change	-0.1000pp

Layer 3 - Threshold and uncertainty contract

Question	Result
Minimum worthwhile effect	1.00pp
Observed point estimate clears MWE?	False
95% CI lower bound clears MWE?	False
Guardrail maximum increase	0.80pp
Guardrail breached?	False

Important distinction: "point estimate exceeds MWE" and "the uncertainty interval establishes at least the MWE" are different claims. M08 should never collapse them into one boolean.

Expected decision language

DO NOT CLAIM A WIN. The point estimate does not clear the 1.0pp minimum worthwhile effect, the CI spans negative and positive effects, and p is far above 0.05. Guardrail is fine, but that alone is not evidence of primary-metric value.

Critical interpretation checks

- State the null hypothesis and alpha separately from the estimated effect.
- Use p-value language conditional on H0; never write that p is the probability H0 is true.
- Use percentage points for absolute rate differences and percent for relative lift.
- Treat the 95% CI as uncertainty around the effect estimate; do not claim a fixed parameter has a 95% posterior probability unless a Bayesian model was actually used.
- A statistically significant result is not automatically business-important. A business-important point estimate is not automatically statistically established.
- Guardrails can block rollout even when the primary metric improves.

Layer 4 - Diagnostic decision tree

- **Counts/labels wrong:** stop. Repair data integrity before calculating rates.
- **Rates wrong:** check numerator/denominator and whether percentages use the full assigned group.
- **pp vs % wrong:** absolute lift = Treatment% - Control%; relative lift = absolute lift / Control%.
- **z/p wrong:** verify pooled SE for the null test and a two-sided p-value.
- **CI wrong:** verify the unpooled SE for the rate difference and keep units in percentage points.
- **Decision too strong:** separately evaluate design/data quality, effect estimate, uncertainty/statistical evidence, MWE and guardrail.
- **Conclusion copied from review:** no pass. Close review and complete changed fresh evidence.

Minimum evidence before pass

Evidence	Pass condition
Saved Gate workbook	Fresh identifiable first attempt exists
Quality block	Counts, duplicate/missing/invalid checks stated
Primary metric	Rates + absolute pp + relative % effect
Uncertainty	95% CI with correct units and interpretation
Hypothesis test	H0, alpha, z, p and reject/fail-to-reject language
Business threshold	Observed point estimate compared with MWE; no overclaim about CI
Guardrail	Both rates + pp change + explicit breach/no-breach
Explain-back	60-90 sec own-words defense without false p/CI/causal claims

Targeted repair routing

Failure pattern	Repair focus
Counts, duplicates, invalid assignment/outcome labels	STAT9 experiment quality + data integrity
Rates/denominators or percentage-point units	STAT1 metric discipline + STAT7 effect size
CI arithmetic or probability-style CI wording	STAT4-STAT5 SE / uncertainty
p-value meaning, H0 or reject/fail-to-reject language	STAT6 hypothesis testing
Observed effect confused with guaranteed MWE	STAT7 business threshold + uncertainty
Guardrail ignored because primary metric improved	STAT7 + STAT9 decision contract
Causal claim exceeds randomized design/evidence	STAT8-STAT9 causal limits/design

Before a fresh retest

- Keep the original failed attempt; do not overwrite it.
- Write the exact competency that failed and the repair evidence produced.
- Close this protected review before starting changed evidence.
- Use a changed scenario or assigned Gate; copying the same controls is not mastery.

Fresh retry and state route

Gate B is remediation-only. Clean B pass ends M08. Failed B -> Review B -> targeted repair -> changed evidence. Gate C only after explicit assignment.

Close this review before the next attempt. Repair only the named weak competency; do not retake by reproducing the same control table.